

SE404.3– Agent Based Systems

Sample Paper

Question 1: Write the correct response in your answer booklet.

[15 marks]

1. What are the two core ideas of the ABM Design Principle?
 - A. Focus on the question at hand and realism
 - B. Simplicity and Focus on the Question at hand (not sure)
 - C. Simplicity and realism
 - D. Realism and any elements desired by the stakeholders

2. What are the three main components of an ABM?
 - A. Behaviors, GIS, Spatial Data
 - B. Networks, Interactions, Environments
 - C. Agents, Networks, Behaviors
 - D. Agents, Environments, Interactions

3. Why is representation in the context of ABM important?
 - A. We must combat taxation without representation.
 - B. No one understands math.
 - C. Representative agents are the key to ABM.
 - D. A good representation helps us better understand the phenomenon we are studying.

4. What are the three tabs of NetLogo?
 - A. Interface, Documents, and Procedures
 - B. Interface, Info, and Code
 - C. User, Reader, and Coder
 - D. Bar Tab, Food Tab, and Drinks Tab

5. What two buttons does almost every NetLogo model have?
 - A. Setup and Go
 - B. Create and Run
 - C. Build World and Build Turtles
 - D. Make Pink Turtles and Party

6. What are the basic mobile agents in NetLogo called?
 - A. Turtles
 - B. Agents
 - C. Humans
 - D. People

7. What is the command to clear the current model of its state? A.
die
B. destroy
C. clear-all
D. clear-drawing
8. How do you start a line that will define a procedure? And what word do you use to indicate you are done with the procedure?
A. to and done
B. two spaces and end
C. procedure and end
D. to and end
9. What is the NetLogo reporter that takes an agentset and returns the members of that agentset (as an agentset) that meet a set of criteria?
A. of
B. with
C. turtle-set
D. patch-set
10. What is the NetLogo command to create different types of agents? A.
type
B. species
C. breed
D. set
11. Analysis of agent-based models is about understanding the relationship of _____ to _____?
A. dependent variables, outputs
B. inputs, independent variables
C. inputs, outputs (not sure)
D. agents, parameters
12. Agent-based modeling is not useful for modeling the behavior of a particular individual because:
A. It only models billions of people.
B. Agents cannot have names.
C. One individual person's behavior may be too specific to capture in an agent-based model.
D. Agents can only be companies.
13. Verification involves comparing the _____ to the _____. (no idea)
A. real world, conceptual model
B. implemented model, conceptual model
C. implemented model, real world
D. my model, your model

14. Validation is the comparison of the _____ to the _____. (no idea)

- A. implemented model, real world
- B. real world, conceptual model
- C. conceptual model, implemented model
- D. first model, second model

15. What is the best level of representation to create a model of wolves and sheep?

- A. Each agent should represent exactly one wolf or exactly one sheep.
- B. It depends on the question you are trying to answer.
- C. Each agent should represent a flock of sheep or pack of wolves.
- D. Each agent should represent one wolf and one sheep.

Question -2 (Insects moving in a grass field)

[25 marks]

- 1) Explain the basic model rules used in modeling insects moving in a grass field (don't know what versions there are 4, this is the 4th one) (8 marks)

Firstly, to enable the grass we ask all the patches to be green. To setup ants we use a predefined number as the number of ants, and we use random x coordinates and y coordinates to place the ants around the world. Then to start the functionalities we make ants move, eat grass, reproduce, and die. To grass to grow back after eaten we use grass-regrow functions. Ants need to die, to enable their death we use an energy factor and slowly decreases it.

To move ants around the world we turn the ants to the right by a random amount of degree from 0 to 360 and make them go forward by 1 unit. Whilst they are moving, we reduce their energy by 1 unit. To kill an ant, we see if the energy of a certain ant is 0, if it is, we make that ant to call die function.

To make an ant to reproduce we simply check if an ant has energy more than a predefined number. If it is true, we ask that ant to hatch another ant with the energy of 50.

To the grass to be eaten we ask that ants who re in a green patch to make it black and we also set that ant's energy level increased by 10 units.

To grow grass there is a rule that only 3% of grass can be grown back. To enable this logic and grow grass back we ask patches to generate a random between 0 and 100 and see if is less than 3. If it is, we make the patch back to green. This will not target the black ones only.

2) Describe an alternative way of effectively displaying the growth of grass? Note: code is not required. (5 marks)

it can be modeled by introducing a grass regrowth rate (a slider) and a function that would color patches of grass with different shades of green (darker means thick grass and light means small amount of grass). When an ant eat grass we reduce the grass amount by the energy factor of that grass and recolor the patch itself to show how much grass it has lost from the patch and ultimately making it black (no grass). Whilst it is happening, we can implement a grass regrow function to monitor the grass and set grass amount for each patch and to monitor the grass amount. This function will also check if a patch a grass amount than a certain amount and readjust the grass amount and calls for grass coloring function.

```
to recolor-grass
  set pcolor scale-color green grass-amount 0 20
end

to regrow-grass
  ask patches [
    set grass-amount grass-amount + grass-regrowth-rate
    if grass-amount > 10 [
      set grass-amount 10
    ]
    recolor-grass
  ]
end
```

3) Imagine that group of bats are now feasting on the insects. Explain how you will model this scenario and how you will monitor the current population of each species? (5 marks)

Firstly, we are required to assign breeds to the model (ants and bats). Advantage of this breeding is that we can set different rules for each breed since they act as different sets of agentsets. We can setup a "catch-ants" function and ask bats to catch ants. In this catch-ants function, we first must set prey to one of the ants currently on a certain patch that the caller itself is on. Then validating after the ant is in fact an agent, we can ask the ant to die. If energy is involved too, we can transfer the ant's energy to the bat.

```
breed [ants ant]
breed [bats bat]

to catch-ants
  let prey one-of ant-here
  if prey != nobody
    [ ask prey [ die ] ]
end
```

to monitor the populations, we can simply use the count functions for ants and bats. Use two monitors one for ants and one for bats. In the ants monitor we can use `count sheep` to monitor and in the bats one does the same but with the respective breed.

4) Now suppose each bat will actively and continuously chase one insect for some time. Discuss the criteria you will use to initially lock what insect to chase and then, how you will make the bat continuously chase it. Also describe the criteria to model the death of insects now? (7 marks)

to enable bats to chase after ants we need to introduce a vision factor to bats, this vision could be extended to be adjusted by the user. We can introduce a function to see If an ant enters the vision area, we can assign that ant as the target and make the bat go directly at the ant. And since ants are also wiggling and going forward this action will look like an active chase. To monitor if the ants are now dead if a chased by a bat, we can introduce a check-bat-catch function that will see if there a direct link from the bat to the ant and ask that ant to die. After that we can set the target back to nobody.

```
ask bats [
  set candidate-ants ants in-cone 120
  ifelse any? candidate-ants [
    set closest-ant min-one-of candidate-ant [distance myself]
    if (target = nobody) [
      set prey-agent closest-ant
      set target prey-agent
      set heading towards prey-agent
    ]
  ]
  fd 1
]
```

in the above code the ants are chosen from the vision cone of bats which is 120 degrees. Then we acquire the closest ant's distance from a bat. After that we set prey as that closest ant and make the bat head towards that ant. If there are no ants, then the bat will simply fly forwards by 1 unit

```
to check-bat-catch
  ask bats [
    if (any? ants-here) [
      ask one-of ants-here [
        ask out-link-neighbors [die]
      ]
      set target nobody
    ]
  ]
end
```

from this function we see if the bat has caught an ant. So, we ask one of the ants to simply die. To determine which ant to die we monitor the links from the bat to ant to see if there a direct link from the bat to ant. If there is a one that ant dies.

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Question-3 (Termites)

[25 marks]

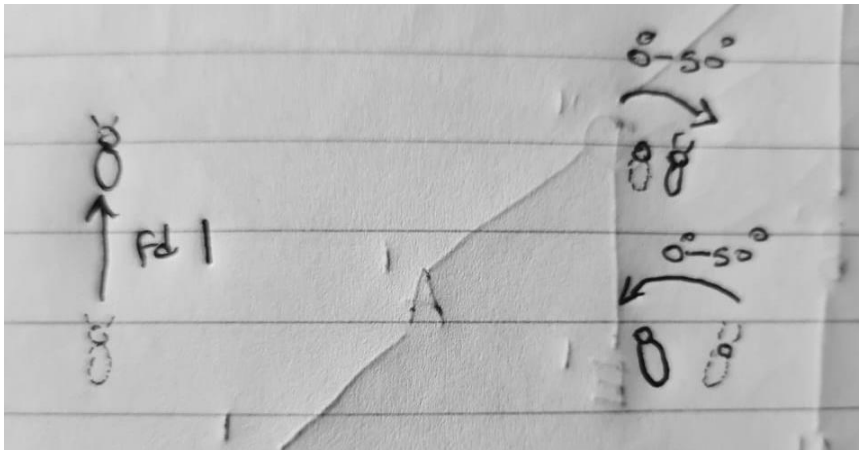
Answer the following questions by referring to the NetLogo code given below.

- 1) State any assumption made in modeling the Termites model. (4 marks)

Each termite starts wandering randomly. If it bumps into a wood chip, it picks the chip up, and continues to wander randomly. When it bumps into another wood chip, it finds a nearby empty space and puts its wood chip down. The termites can wood from the pile too it is not covered. No predefined patch for wood to pile. No leader termite

- 2) Describe the wiggle procedure (6 marks)

It is used to make the termites move all over the place. It is only triggered when the patch is black. It makes the termites go forward by 1 unit and turn to the right to a random amount of degree between 0 and 50. It does this same thing but for the left direction.



- 3) Describe put-down-chip turtle procedure (7 marks)

This procedure makes the termite to put the wood chip down in a black patch. First it checks whether a certain patch is black or not. If it black, then the patch set to yellow (to imitate that a termite has picked up wood chip and put it down). After this action, the termite's color set to white and calls for the "get-away" function. As discussed above if a certain chip is not black then the termite is rotated to the right direction by random amount of degrees from 0 to 360 and move forward by 1 amount and calls the "put-down-chip" function.

According to the above mentioned "get-away" function is triggered when a termite is on a black patch. In that function a termite is rotated to the right direction by random amount degrees from 0 to 360 and move forward by 20 units. And if the termite is still on a patch that is not black (yellow, assume this as a wood pile) then the "get-away" function will be called again until the termite is moved out of the yellow patch.

- 4) Explain the code given below with a brief introduction to the model. (8 marks)

The setup function – it starts off with clear-all command, which clears all agents (turtles, patches, observers, and links). Then it goes to set a shape to the turtles which is "bug". After that it goes to ask patches to grow. First it expects a user input (a number between 0 and 100) as the density, then it generates a random float value between 0 and 100. After that it checks if the user input is greater than the randomly generated value. Assume the user gave 20 as the input then when the patches will check the random number is less than 20 and make a particular patch yellow. Note that if the user gives 100 as the user input the whole world will be yellow since any randomly generated number between 0 and 100 will be less than 100.

100 is always less than 100. This is implemented to give the user the sensation that whatever the number she enters is the actual percentage of wood chips there will be. After the patches it goes to creating turtles. The number of turtles is determined by the user. Color of the turtle will be white, and the turtles will be placed around the world according to the random x-coordinates and random y-coordinates generated. Size of the turtle will be 5 units.

The search-for-chip function – it checks if a certain patch is yellow or not. If it is yellow, then the same patch will be converted to yellow (to imitate that the yellow wood chip has been picked by a termite) and the termite will be moved forward by 20 units whilst its color converted to orange to denote that the termite is occupied. As discussed above if the patch is black (not yellow) the termites will call wiggle and search-for-chip respectively.

The find-new-pile function – if a patch is not yellow (black) the termites will call wiggle function and find-new-pile function respectively

Others are already discussed

The bottom line of this code is that some patches are colored to symbolize wood chips. Termites will wiggle around the world and if they move to a yellow patch the patch will be marked black and the termites will be moved according to the find-new-pile function until they end up in a black patch and executes the put-down-chip function

```
to setup
  clear-all
  set-default-shape turtles "bug"
  ask patches
  [ if random-float 100 < density
    [ set pcolor yellow ] ]
  create-turtles number [
    set color white
    setxy random-xcor random-ycor
    set size 5
  ]
end

to go
  search-for-chip
  find-new-pile
  put-down-chip
end

to search-for-chip
  ifelse pcolor = yellow
  [ set pcolor black
    set color orange
    fd 20 ]
  [ wiggle
    search-for-chip ]
end
```

```
to find-new-pile
  if pcolor != yellow
  [ wiggle
    find-new-pile ]
end

to put-down-chip
  ifelse pcolor = black
  [ set pcolor yellow
    set color white
    get-away ]
  [ rt random 360
    fd 1
    put-down-chip ]
end

to get-away
  rt random 360
  fd 20
  if pcolor != black
  [ get-away ]
end

to wiggle
  fd 1
  rt random 50
  lt random 50
end
```

Question-4 (Mousetraps)

[35 marks]

- 1) State an assumption made in the model implementation below. (5 marks)

Mouse traps will trigger one after other and each turn is an exponential growth of the previous one. It is a metaphor for nuclear fission. Even if a sustained chain reaction occurs, not every mousetrap gets triggered. The shape of reaction propagation is almost always a circle.

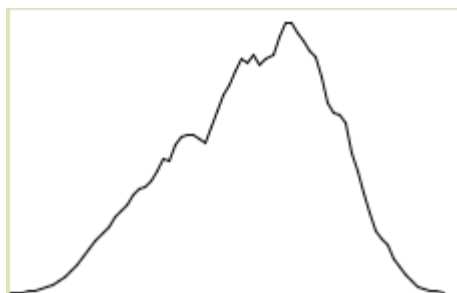
- 2) Describe in general, what you will observe if you are to run the below program. (5 marks)

if Balls in the air is plotted it would go up exponentially and comes down exponentially. The traps triggered plot always looks like an "s" curve, exponentially rises, and stops rising at a certain amount.

- 3) Even if a sustained chain reaction occurs, not every mousetrap gets triggered. Why? (5 marks)

The way this is set the ball will rotate to the right direction by (0 – 360 degrees) and move forward by a user input. By doing this we cannot guarantee that the balls will only hit the untriggered mouse traps, sometimes it can hit an already triggered mousetrap. So, if it hits an already triggered mousetrap nothing happens. Simply the balls do not know where the untriggered mousetraps are hence not mousetraps will not be triggered

- 4) What will be the shape if "Balls in the air" is plotted? Why do you think it's shaped that way? (5 marks)



It will look like the above picture. The ascending occurs to the exponential growth of the balls that are being hatched. The ball lands on a trap, the trap triggers, and a second ball flies into the air. The first ball also bounces into the air again, so now there are two balls in the air. Each of those two balls triggers another trap, so there are four balls in the air. However, the untriggered traps will now be exponentially reduces, so there will less and less amount of balls to be hatched. That is the reason for the steady decrease.

- 5) Analyze the given code and explain it in a detailed manner. You need to include a brief introduction to the model in your explanation. (15 marks)

this model can be used to demonstrate nuclear fission. Light blue squares represent untriggered mousetraps. Red squares represent triggered mousetraps. When a ball lands on a light blue square, the square turns red, a new ball appears, and both balls travel a random distance in a random direction

the setup function – it starts off with clear-all command, which clears all agents (turtles, patches, observers, and links). Then it sets the traps-triggered to 0 as this this the initial round. Then it sets patches to blue + 3 (blue is 105 so 108 in this case is a lighter shade of blue). Then it sets the turtle shape to a circle and creates a single turtle with 1.5 size then it resets the ticks which Resets the tick counter to zero, sets up all plots, then updates all plots (so that the initial state of the world is plotted).

The go function – it checks if there are no turtles exists in the world if not then the reaction stops. Then it checks if the patch color is red it does this to check if a ball hit the wall or land on a triggered mousetrap, so it can make that turtle dies. If the patch is not red, then the patch will be converted to red color and sets the global traps-triggered to traps-triggered + 1 and creates 1 another new ball. As soon as this ball created it calls for the move function. Then the ball that helps to hatch the new ball also calls for move function. So, the new ball and the old ball both will move.

The move function – it rotates the turtle to the right direction by random float degrees from 0 to 360 and moves the rotated turtle to forward by a random fraction between 0 and the max distance set by the user.

```
globals [traps-triggered]

to setup
  clear-all
  set traps-triggered 0
  ask patches
  [ set pcolor blue + 3 ]
  set-default-shape turtles "circle"
  create-turtles 1
  [ set color white
    set size 1.5 ]
  reset-ticks
end

to go
  if not any? turtles
  [ stop ]
  ask turtles
  [
    ifelse pcolor = red
    [ die ]
    [ set pcolor red
      set traps-triggered traps-triggered + 1
      hatch 1
        [ move ]
      move ] ]
  tick
end

to move
  rt random-float 360
  fd random-float max-distance
end
```